

Exploring Slope and Intercept in Linear Graphs

SimuPhysics Slope Dynamics Visualizer · SimuPhysics

Years 7-8	~30 minutes	Mathematics · Algebra & Functions
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Simulation: <https://simuphysics.com/view/mathematics-slope-dynamics-visualizer-1762145798110.html>

Curriculum links: AC9M8A02 · AC9M7A04

Learning intentions

- I can describe how changing the slope of a line affects its steepness and direction.
- I can describe how changing the y-intercept affects where a line crosses the y-axis.
- I can predict how a line will change before I move a slider, and check my prediction.

Getting started

Open the SimuPhysics Slope Dynamics Visualizer. Before touching any sliders, note the starting equation and what the line looks like. Then adjust the slope slider first, leaving the intercept slider unchanged.

Tasks

1. Before moving any sliders, predict what will happen to the line if you increase the slope value. Then move the slope slider and check — were you right?

2. Set the slope to a negative value. How is the line different from when the slope was positive?

3. Try four different slope values and record the steepness and direction of the line for each.

Slope value	Steepness (steep/gentle)	Direction (up/down)

4. Reset the slope and now adjust only the intercept slider. What part of the graph moves, and what stays the same?

5. Set the slope to 0. Describe what the line looks like and why this makes sense given what slope represents.

6. Explain, in your own words, what the slope and the intercept each tell you about a linear graph. Use the equation shown on screen to support your answer.

7. Using the equation shown for your final line, calculate the y-value when $x = 5$, then check this against the graph.

Working:

Extension (optional)

Two lines have the same slope but different intercepts. Use the simulation to set this up and explain what this means about the two lines geometrically. Think about whether they ever meet.
